PARAMETER	VALUE
Part Number	517143700101,G,2
Description	AIR COMPRESSOR
Modification Description	4Cy eCompressor changes -eComp without elbow.
Track & Trace	Yes
Critical to Emission / Critical to Safety	CTE

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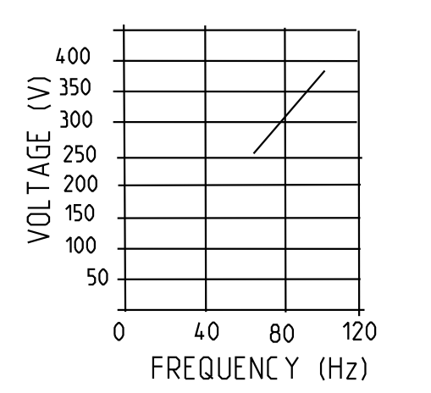
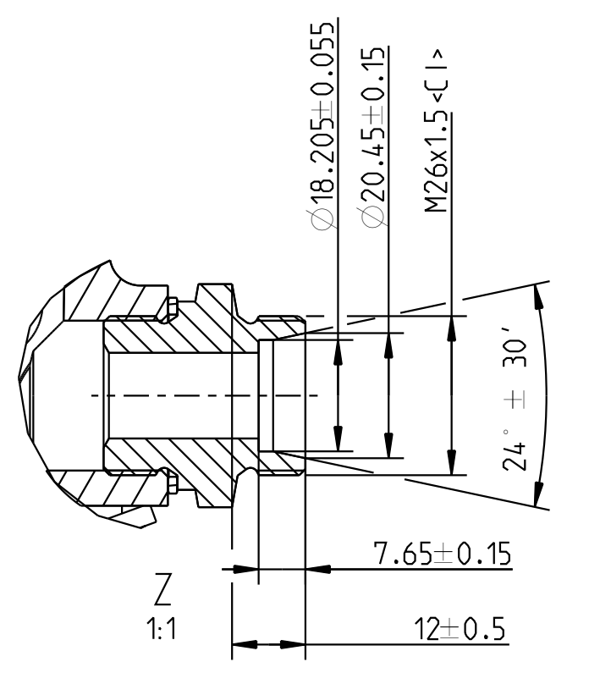
Do not scale the drawing.	Unless otherwise stated. 1. All dimns in mm. 2. Drg drawn to first angle projection. 3. For supplier drg release, refer angle of projection on supplier drg.	Nominal Dimension	Tolerance in µm(microns) Refer IS919-2
If in doubt, ask.		-	-

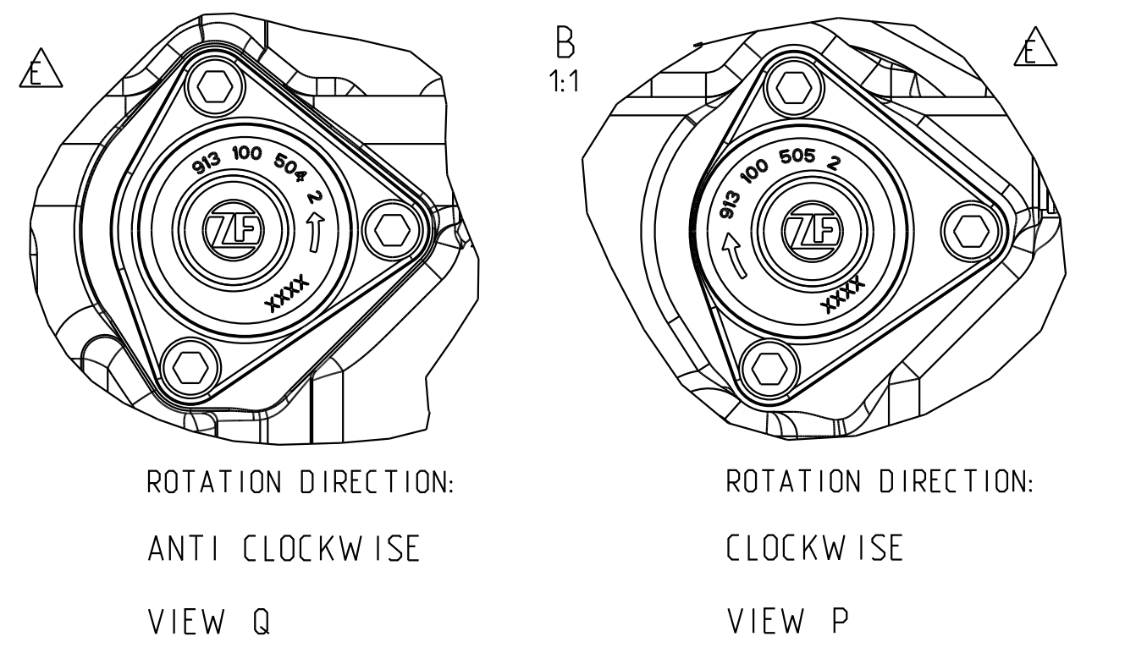
	TECHNICAL DATA				
	COMPRESSOR	E-MOTOR	TEMPERATURE SENSOR PT1000(G)	ELECTRICAL FAN (F)	OIL PRESSURE SWITCH (H&J)
TYPE OF CURRENT	------	ALTERNATING CURRENT	DIRECT CURRENT		
RATED VOLTAGE	------	380V	-----	24V	24V
RATED CURRENT	------	7.8A	0.1--0.3mA	1.44A	1A max.
MAX. CURRENT	------	18.4A			
DUTY CYCLE IEC 60034-1 S3 U:380V P1:10 bar Pamb:1013mbar	ALLOWABLE AIR CONUMPTION 100 lpm 1500 rpm < 25% 1250 rpm < 30%		100%	100% PERMISSIBLE (MAY NOT WORK WHEN TEMPERATURE IS LOWER -20°C	100% PERMISSIBLE
DEGREE OF PROTECTION	IP67				
OHM RESISTANCE	-----	2.18Ω±2.5%	R=839.14Ω at -40°C R=1081.43Ω at +20°C R=1158.88Ω at +40°C R=1350.49Ω at +90°C	------	CONTACT RESISTANCE MAX 50 mΩ FOR NEW
AMBIENT TEMPERATURE WITH FUNCTION	-25°C----+60°C				
NUMBER OF POLES	------	8	------	------	------

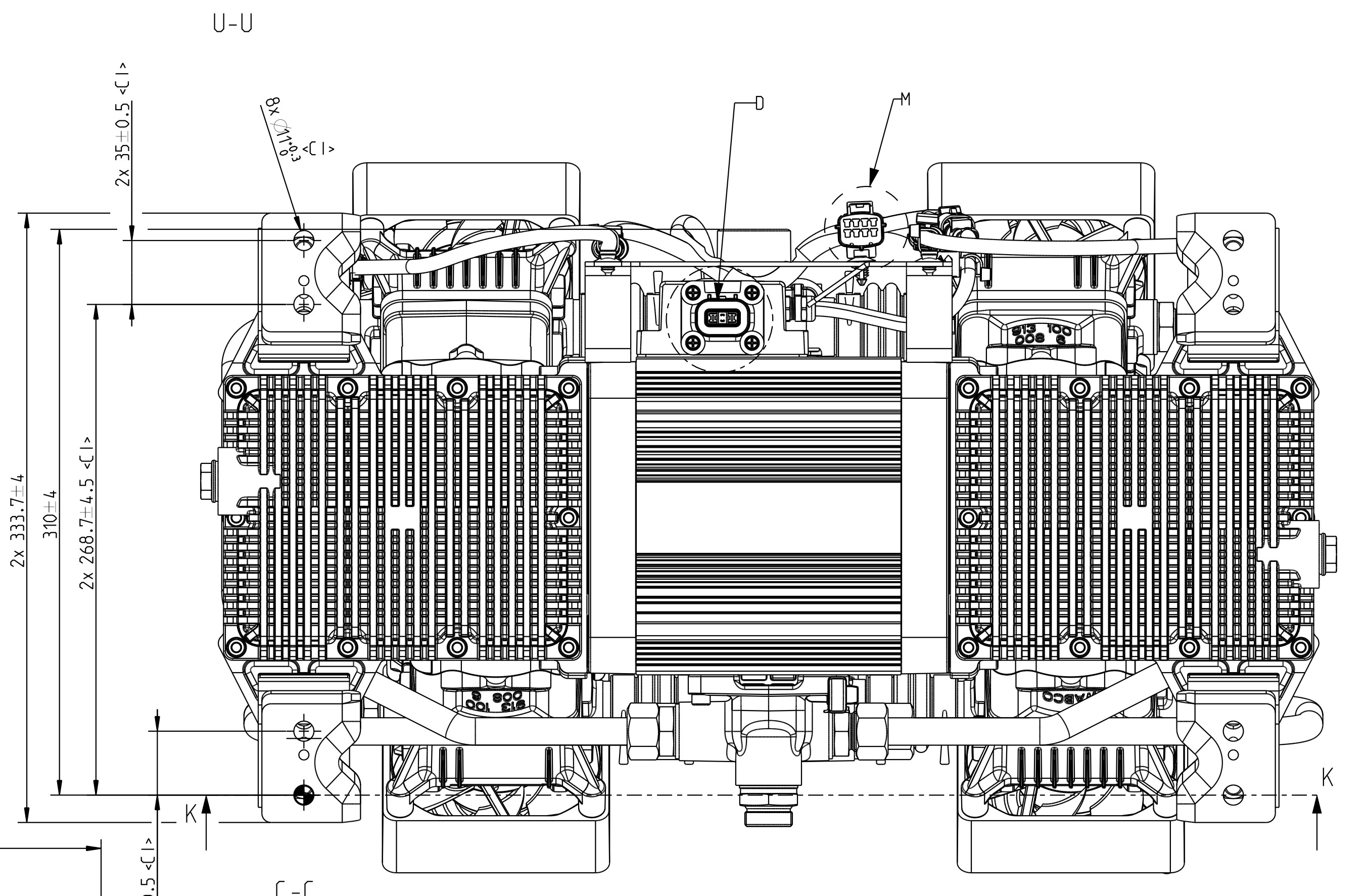
VOLTAGE,FREQUENCY vs RPM MOTOR DATA		
RPM	VOLTAGE (V)	FREQUENCY (Hz)
1000	253	67
1250	317	83
1500	380	100

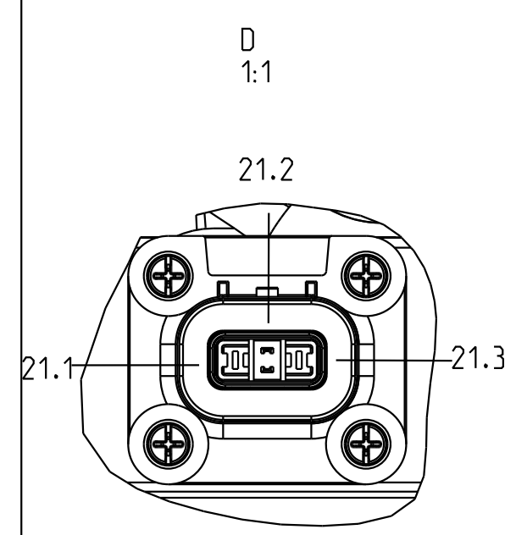
VOLTAGE VS FREQUENCY MOTOR DATA







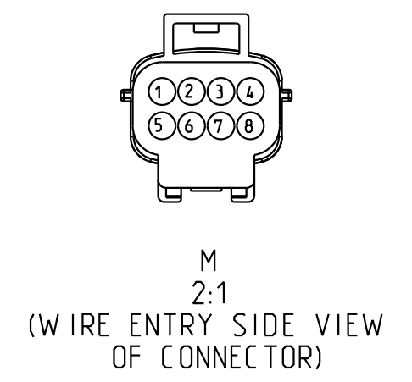




<CI> CUSTOMER INTERFACE	10

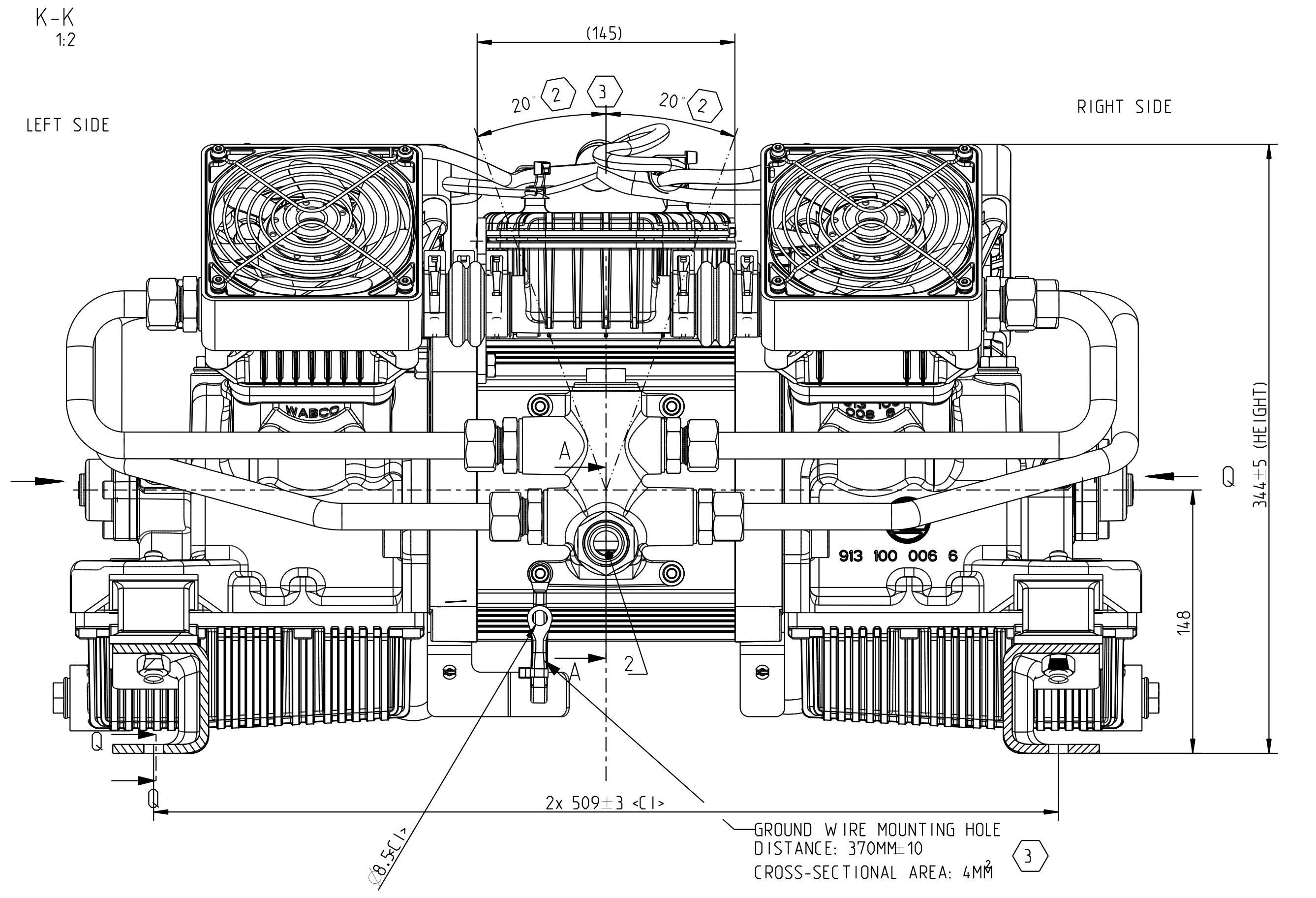
TE SOCKET			
HVA280 3P HEADER ASSY 3MM²(SPLIT) CODE B			
2-2310541-2 <CI>	HVA280 3P SPLIT HERAER ASSY KEY B		1
2103321-2	3P INNER HSG, SPLIT HDR, KEY B, HVA280		1
2-2141600-2	MCP2.8MM TERMINAL SILVER PLATED		3
HOLE CODING	21.1	21.2	21.3
DOWN-LEAD DEFINE	U	V	W
LEAD COLOR	YELLOW	GREEN	RED

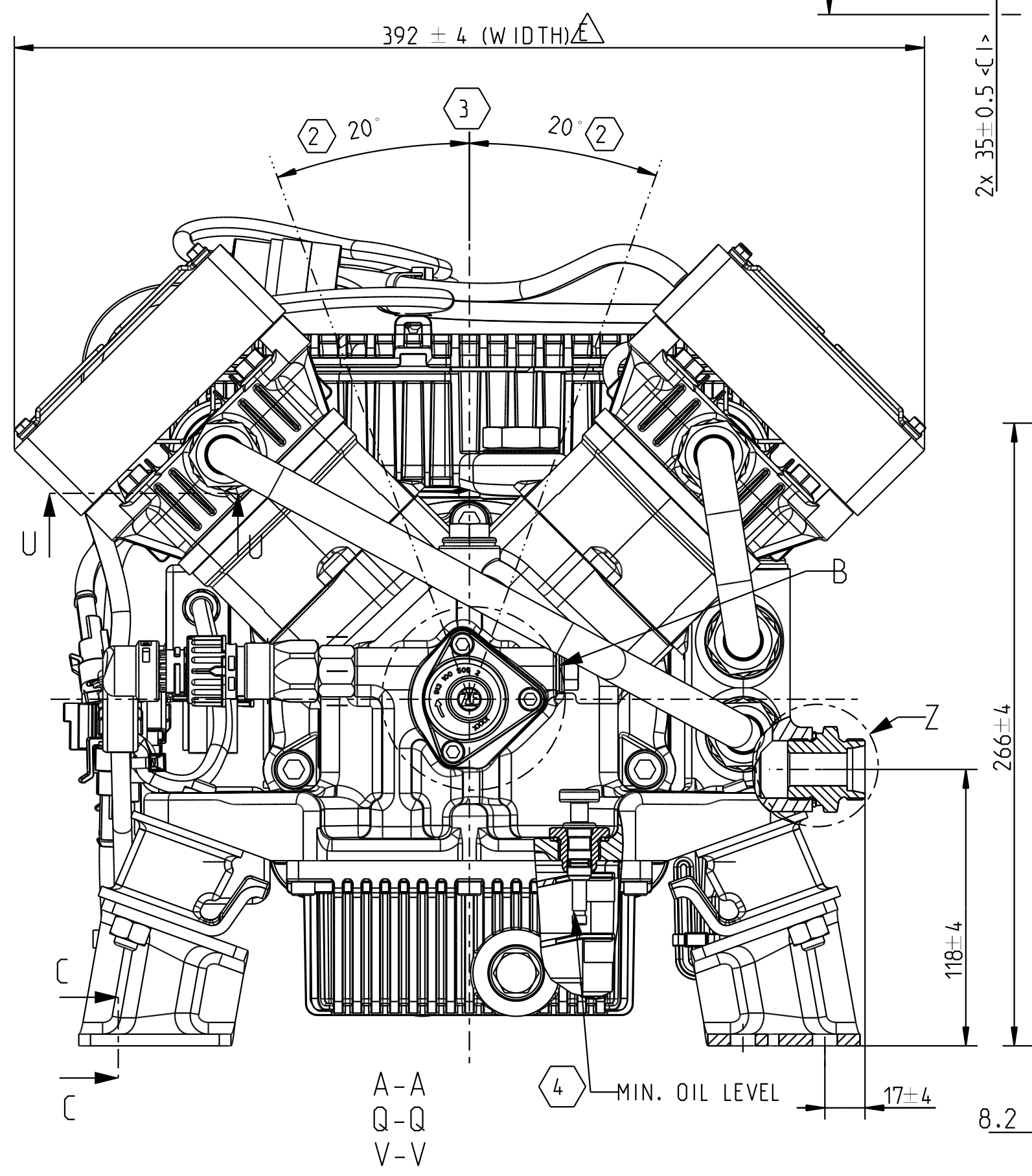
MATCHING TE PLUG		
HVA280 3PHM PLUG ASSY 3x2.5MM² COROPLAST CODE B		
4-2103015-2	PLUG SUBASSY,B,HVA280-3PHM	1
2103306-1	INNER HOUSING,PLUG,HVA280-3PXM,XE	1
1-968851-3	AMP MCP 2.8, CONTACT	3
1-2103157-2	SHIELD, PLUG, SIZE 2, AG, HVA280	2
1587723-2	INNER FERRULE, SIZE B, HVA280-	1
1587724-2	OUTER FERRULE, SIZE B, HVA280	1
2103153-1	SPACER, HVA280-2PHM	1
2103154-2	CABLE SEAL, SIZE 2, HVA280	1
2103155-2	COLLET, SIZE 2, HVA280	1
2103013-2	CABLE SEAL RETAINER, SIZE 2, BK, HVA280	1

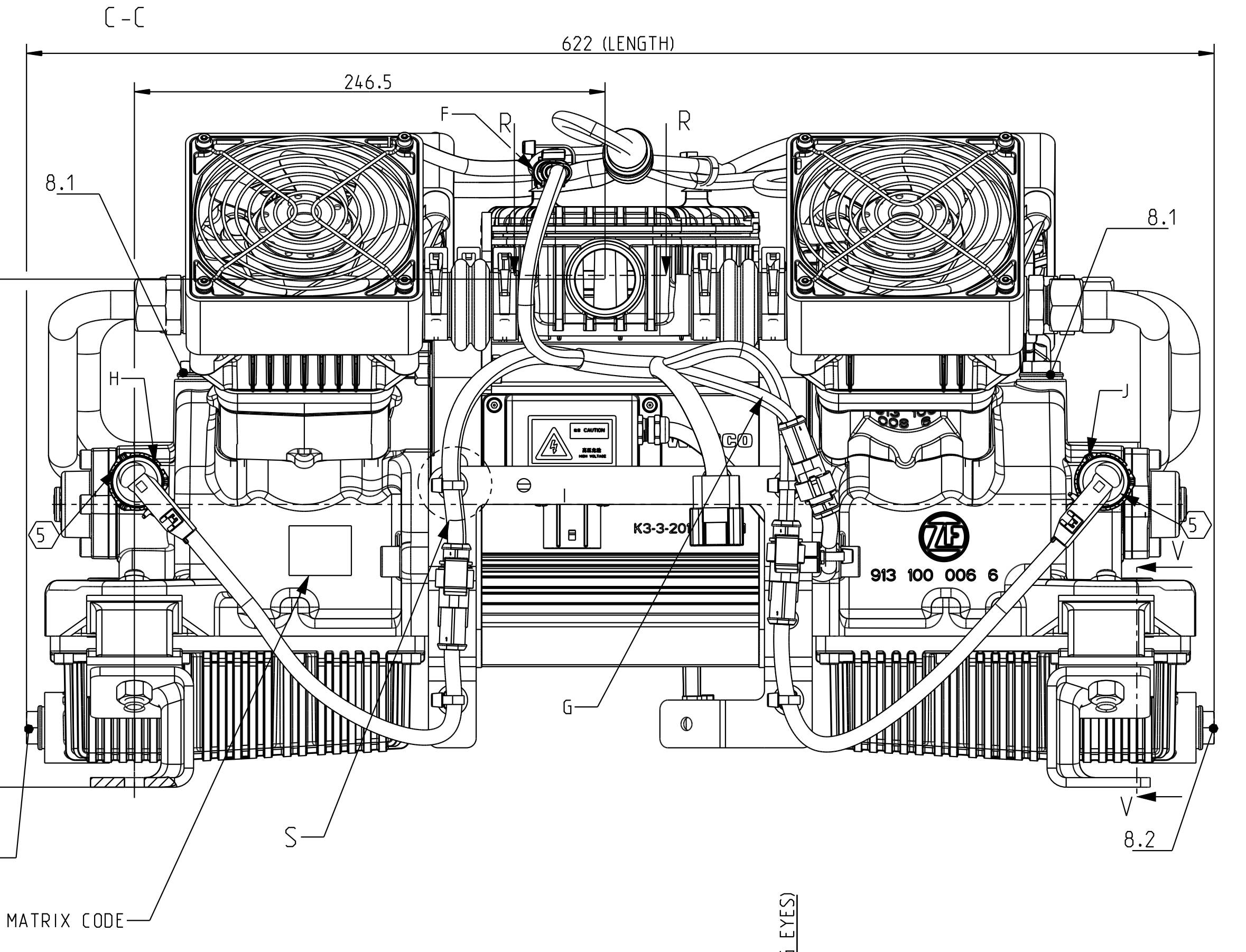


M
2:1
(WIRE ENTRY SIDE VIEW OF CONNECTOR)

8P CONNECTOR DETAILS		
E-COMP CONNECTOR <CI>		VEHICLE CONNECTOR
TE 2307645-1 (SEE DETAIL M)		TE 2307642-1
PN. No	DESCRIPTION	DETAILS
1	FAN +	F
2	FAN -	
3	MOTOR TEMP S +	G
4	MOTOR TEMP S -	
5	PR SW1 INP	H
6	PR SW1 OUT	
7	PR SW2 INP	J
8	PR SW2 OUT	







① DATA MATRIX CODE

- WARNING HIGH VOLTAGE

DEGREE OF PROTECTION: REFER PRODUCT SPECIFICAITON DOC. 035

ELECTRICAL GROUNDING NECESSARY

SWEPT VOLUME: 520 ccm

RATED WORKING SPEED (DEP. ON MOTOR SPEED): n= 1500 min^{-1}

MAX. WORKING SPEED : n_{max}= 1500 min^{-1}

MIN. WORKING SPEED : n_{min}= 1000 min^{-1}

MAX. WORKING PRESSURE: P1 = 10 bar

MAX. PRESSURE AT DISCHARGE PORT: P = 12.5 bar

HORIZONTAL MOUNTING POSITION

LUBRICATION: INTERNAL LUBRICATION SYSTEM WITH REPLENISHMENT

OIL VOLUME: 1.0 LITER FOR EACH SIDE, TOTAL 2.0 LITER

OIL TYPE: 10W40 FULL SYNTHETIC OIL

MOUNTING SPACE REQUIREMENT: LENGTH x WIDTH x HEIGHT: 732mm x 502mm x 399mm

NOTE:

MOUNTING HOLE NEED TO BE OBLONG HOLES, DAMPER DEFLECTION AFTER ASSEMBLY IS NOT ALLOWED.

CABLE CONNECTOR POSITION ON VEHICLE NEED AUTHORIZED BY ZF CVS

② MAX. PERMISSIBLE TILT ANGLE DURING CONTINUOUS WORKING AT VEHICLE CONDITION

◐ DATUM MOUNTING HOLE

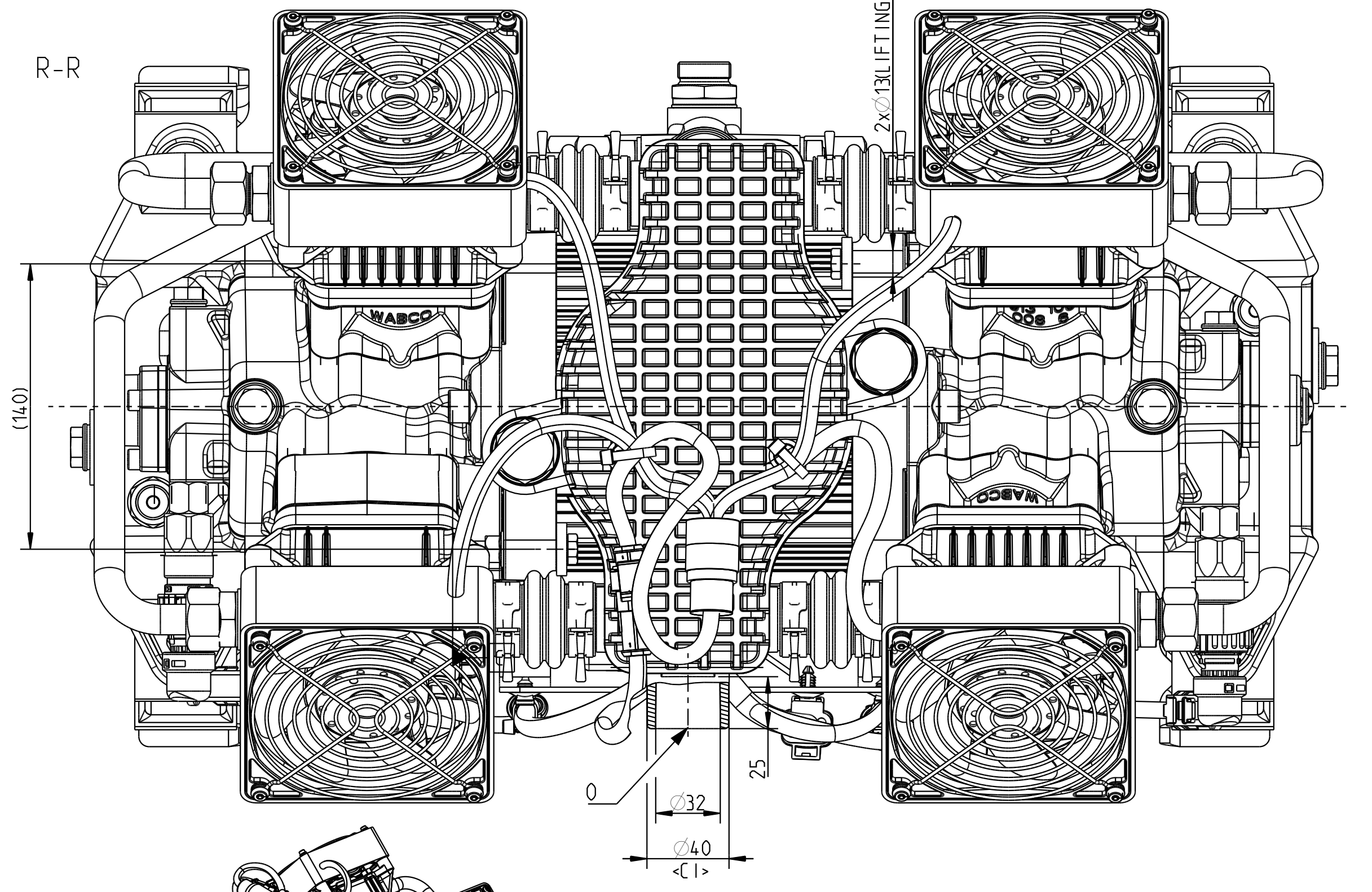
③ ONLY FOR CUSTOMER INFORMATION

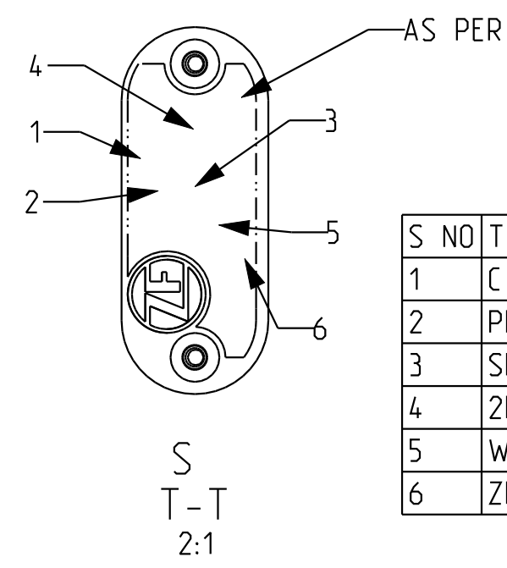
④ DIPSTICK FOR OIL LEVEL INDICATION

⑤ OIL PRESSURE SWITCH - SWITCHING RANGE : 1.3±0.4 bar

① DATA MATRIX CODE:

SIZE:MIN.10x10 / MAX.25x25
ADDITIONAL QUIET ZONE OF 2mm PER SIDE

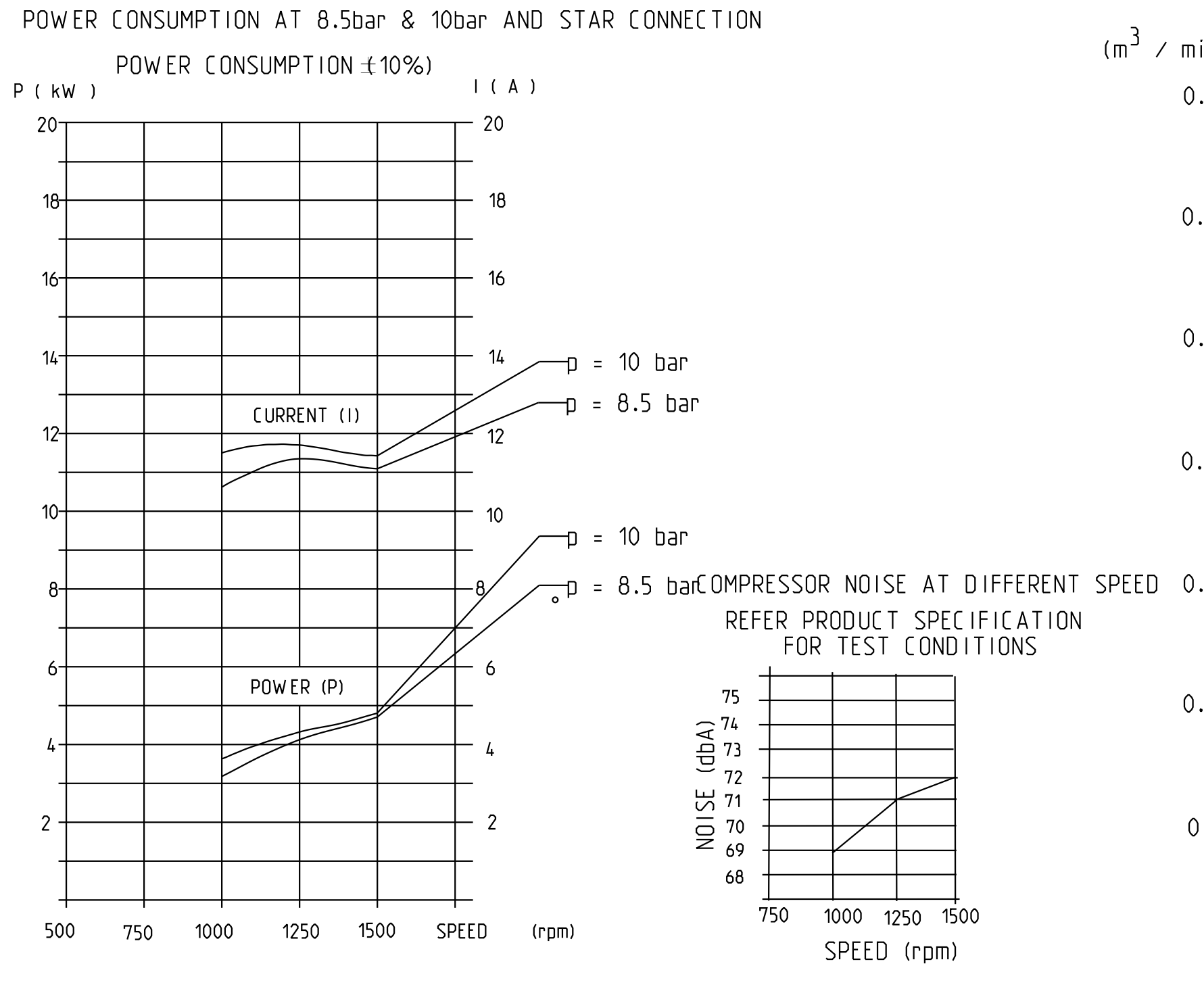




S NO	TEXT
1	CUSTOMER PART NUMBER
2	PRI
3	SERIAL NUMBER
4	2D DATA MATRIX
5	WEEK OF MANUFACTURE (WW YY)
6	ZF PART NUMBER

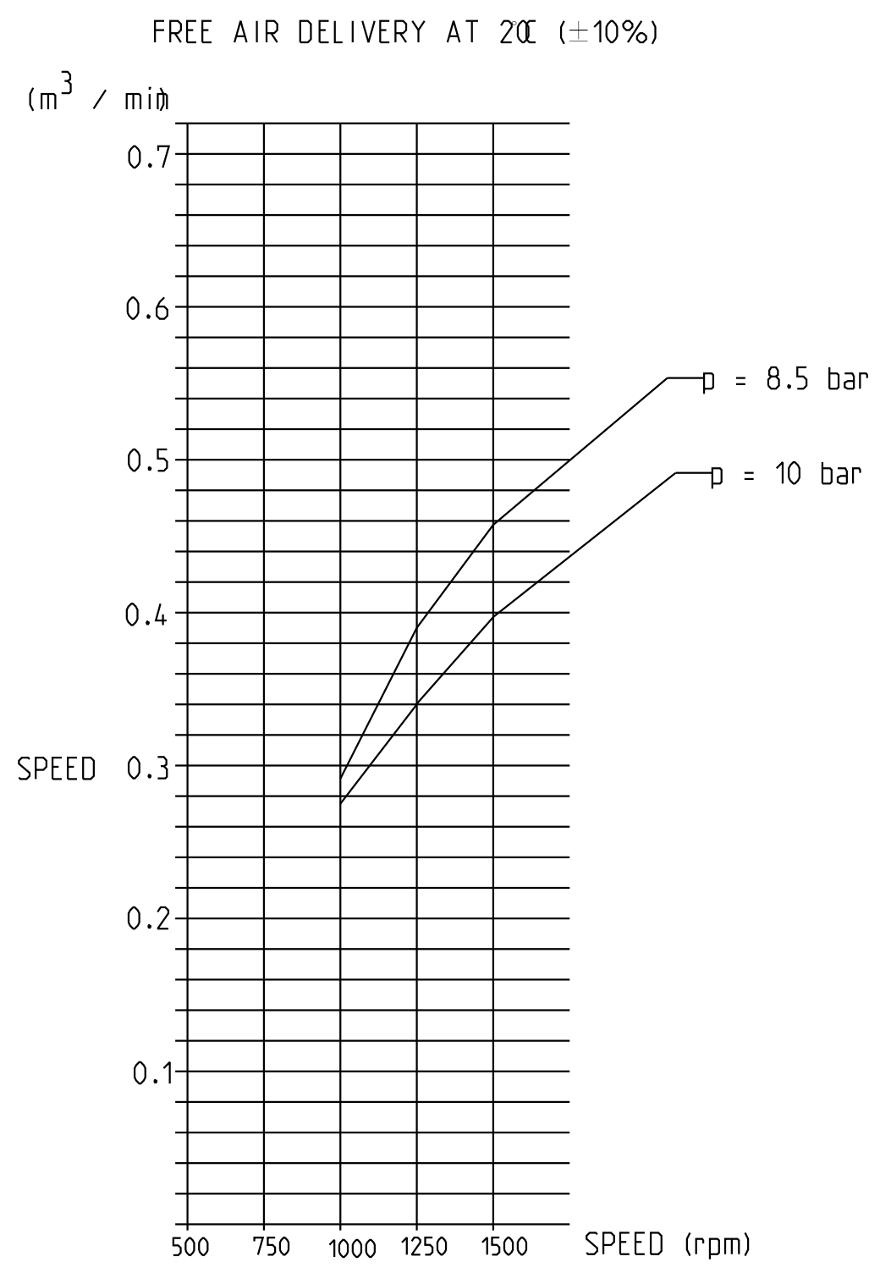
GREEN REPORT RELEASE NO. VT/TR/101/24

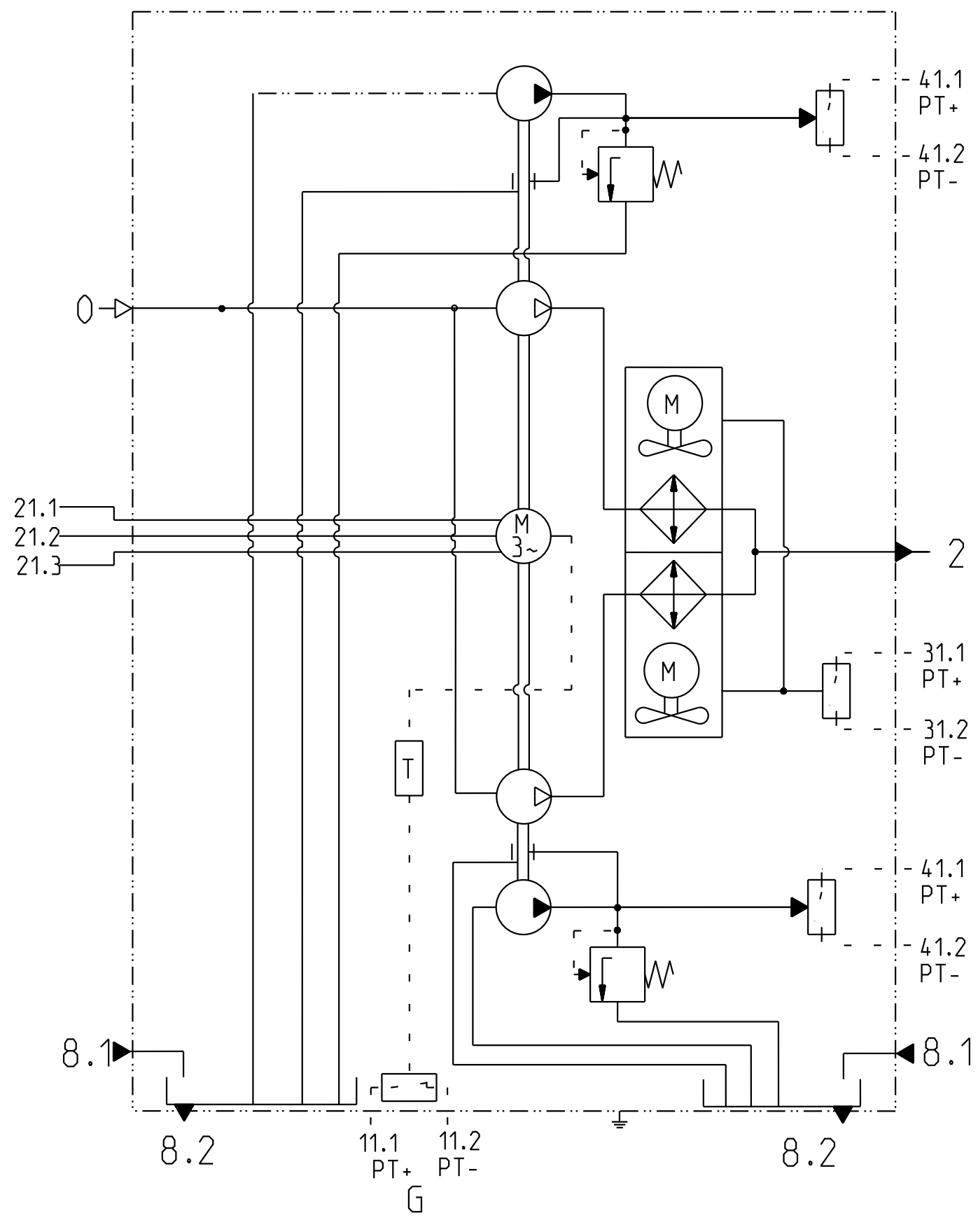
POWER CONSUMPTION AT 8.5bar & 10bar AND STAR CONNECTION

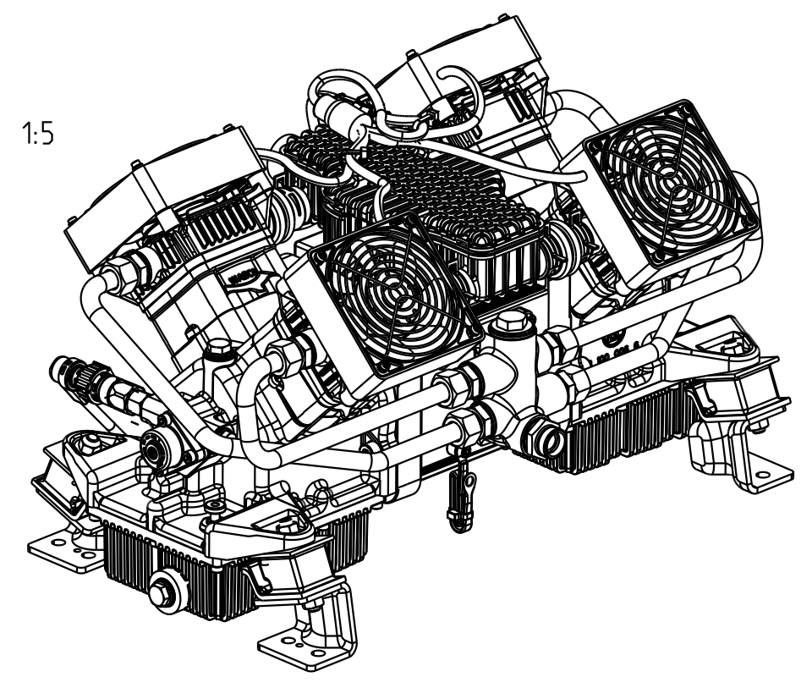


DUE TO RESONANCE COMPRESSOR SPEED RANGE 1050-1150rpm IS NOT RECOMMENDED

FREE AIR DELIVERY AT 20C (±10%)







TORQUE REQUIREMENT
(FOR CUSTOMER INSTALLATION)

O = INLET PORT-
CLAMP RING (SCREW TYPE) M= 5 ± 1 Nm

2 = DISCHARGE PORT M= 60 ± 7 Nm

8.1 =OIL REFILL M = 40 ± 4 Nm

8.2 =OIL DRAIN M = 40 ± 2 Nm

General Specification: ISO 8015, JED-334-1, Size ISO 14405 LP					
Further Technical Data: 9131000010					
Doc. Code:035		Sheet: 1 To 16			
General Tolerances JED-261					
Class 1)		Range of nominal dimensions in mm ≤ 50	> 50 ≤ 180	> 180 ≤ 400	> 400
Fine		±0.5	±1	±1.5	±2
Medium	X	±1	±2	±3	±4
Coarse		±2	±3.5	±5	±6.5
Tapped Holes acc.					
1) Tolerance Class Applied Crossmarked					

Angular tolerance: ±3°

Copyright ZF®	
Date	Signature
Drawn	
2024-09-23	GOPAL
Checked	
2024-10-08	Ni
Dimensions in mm	

COMPRESSOR;E-DRIVEN

Mass	Scale	Material No.	Date of first issue:2021-11-15		
51.55 kg	1:2 (1:5,1:1,2:1)	913 100 001 0	Doc.Code	Language	Sheet
			005	EN	1/1
Size	CAD System				
A 0	CREO				

ECN-No.	Revision	Techn. Resp.	Replacement for
516725	4 E	5103	-

100mm

For Bar Codes System of Supplier Parts, refer TS 11879
For Numbering of Aggregates Manufactured in-house, refer TS 11842
For List of Parts having Recall & Traceability Requirements, refer TS 10814

For all Vehicle Components & Assemblies.
Tata Group Mark & Part No. to be marked as per TS 10806

For M1 & N1 class of vehicles, applicable standards are:
TS11414,TS11418,TS11419 & TS11420

Material description,size,spec,std No.

Proprietary Part: Refer Supplier Drawing

Classification of characteristics:
- ● *Critical.* ▲ *Key Instruction*
- ◐ *Major.* *(None) Minor*
- ● *Applies to upper limit*
- ● *Applies to lower limit*

Date	Mod. No.	Sl No.	Zone	Modification	Chkd	Appd
12/12 2024	-	g	-	4Cy eCompressor changes -eComp without elbow.	APA	PRA
18/11 2024	22CU V40 014	f	-	E COMPRESSOR ASSY. UPDATED WITH CHILD PART LOCALIZATION: 1. RESONATOR 2. OIL SUMP 3. DISCHARGE PIPE 4.CY. COVER 5.DISCHARGE TANK 6.C CLASS PARTS 7.CYLINDER LINER 8.CRANK SHAFT 9.HOLLOW SCREW 10.HOSE BELLOW 11.WASHER & 12.CASTINGS 13.OIL PUMP	APA	PRA
04/12 2023	15PM 440 109	e	L8	ELECTRICAL CONNECTOR DETAILS ADDED	APA	MVS
29/09 2023	15PM 440 104	d	- - - - L7	DRAWING REVISED AS PER SUPPLIER'S LATEST REVISION NO. 'D'. PRESSURE SWITCH VIEW UPDATED CAD UPDATED NOTE ADDED FOR BARCODE DETAILS	APA	MVS
10/09 2022	12PP 287 054	c	-	CAD UPDATED	APA	MVS
27/08 2022	15PM 440 069	b	-	SUPPLIER DRAWING REDRAWN	APA	MVS
17/06 2022	14PM 222 068	a	-	CAD UPDATED	APA	MVS
19/12 2022	20PM 312 000	NR	-	NEW RELEASE	-	-

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2024	SIGN	DATE
Drawn	RKG	13/12
Chkd	APA	13/12
Appd	PRA	13/12

Scale: 1:1

Envelope Dimensions	*Final mass in kg.*
356.07 x 389.82 x 622.00	51.460

Product/Group No. 9/12 EV PMeSEVA BUS.

Surface protection as per standard	-
Tolerance as per standard	-
Replaces Drg	-
Reference Drg No.	-
Opposite Hand Drg/Part No.	-

TATA MOTORS LIMITED
ERC - PUNE

Drg/Part Designation

AIR COMPRESSOR
-

Drg/Part No.

5171 4370 01 01

Sheet No....1.... of ....2.... Sheets

ⓓ REFER DRAWING - 5080 4370 00 08 FOR BARCODE DETAILS.

ⓔ

ELECTRICAL CONNECTOR DETAILS				
E COMP. CONNECTOR	SUPPLIER NAME	TERMINAL PLATING	IP RATING	EXPECTED I/F CONNECTOR FROM WIRING HARNESS
2307645-1	TYCO	TIN	IP67	2307642-1

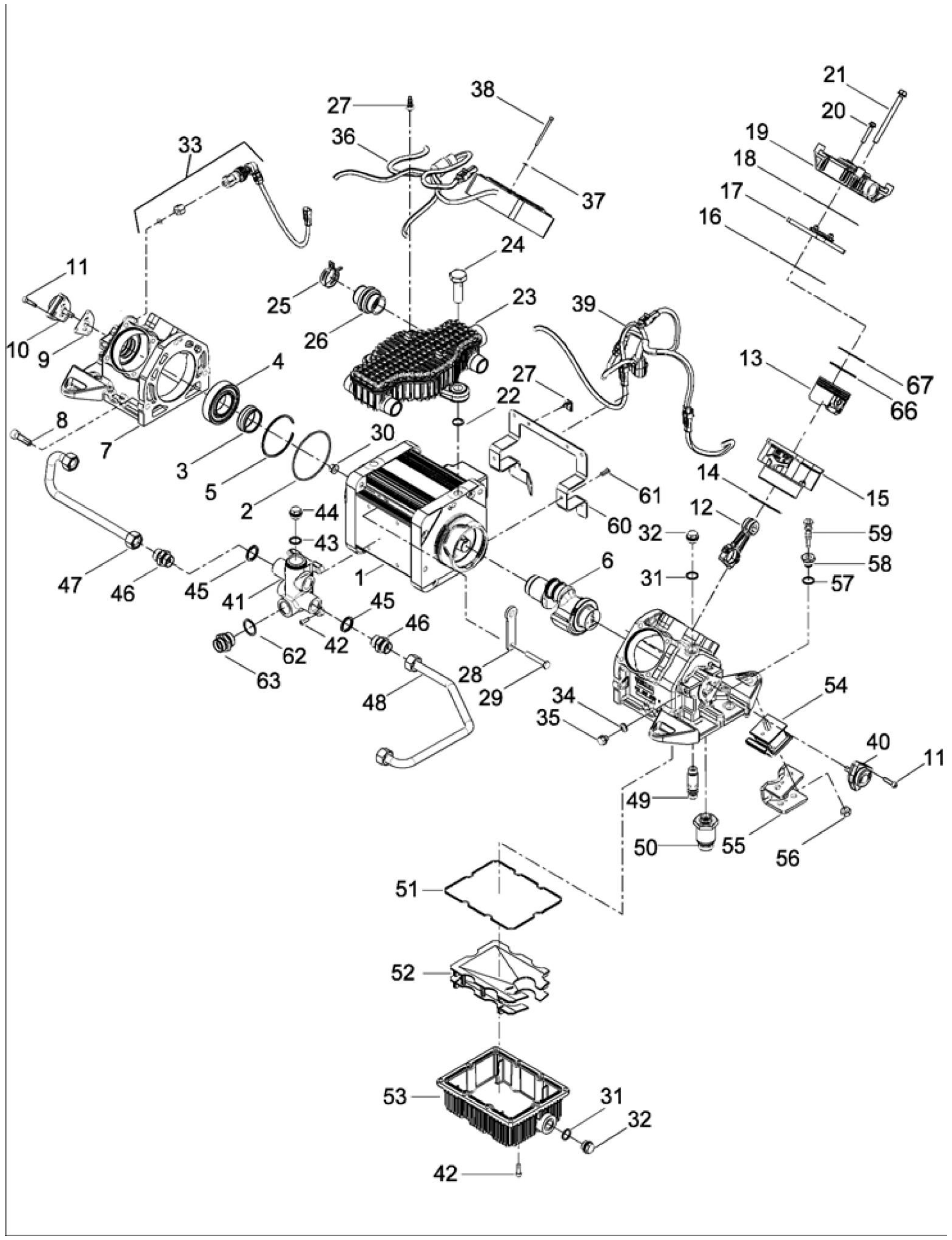
For explanation of drawing format contents, refer to TS11103
8845 4130 70 66 PCF- 071(R4) 6/2011

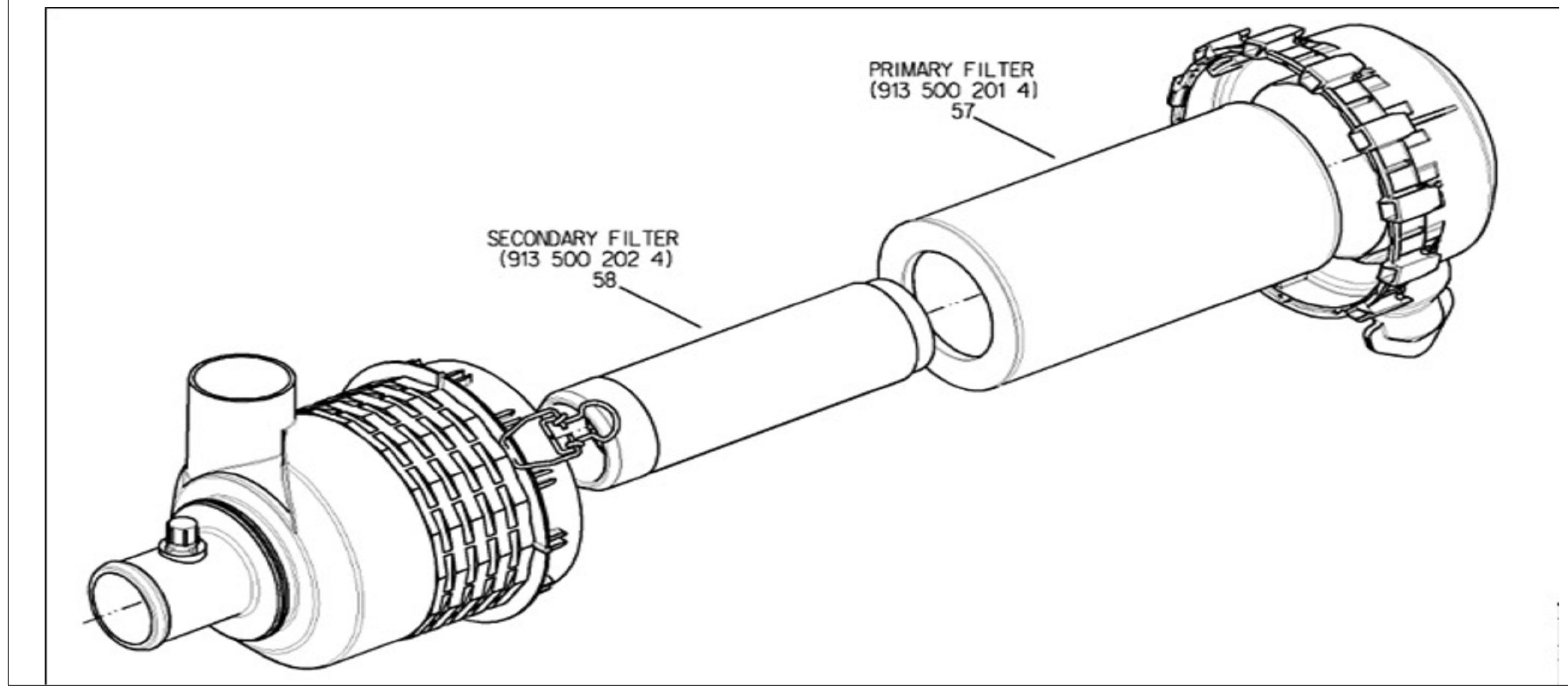
Original drawing size A0

E-Comp 2.0_Indian Variant (913 100 001 0) Service Kit Details

ⓓ

Exploded View





Kit Sl.no	Kit Part No	Kit Name	Sl. No	Item No.	Description	Qty	Customer Part No
1	913 100 930 2	Motor Crankshaft Assy Kit	1	1	Electric Motor	1	517143990101
			2	2	O-Ring - Crankcase	2	
			3	8	Cheese head screw M10x35	6	
			4	30	Hexagon nut M8	2	
			5	29	Hexagon head screw M8x50	2	
			6	28	Bracket	2	
			7	27	Cable tie	4	
			8	14	O-Ring - Liner	4	
			9	16	Inlet Valve	4	
			10	18	Gasket for Head	4	
			11	20	Cheese head screw ISO14580-M6x35	4	
			12	21	Hexagon Flange Bolt M8X100	16	
			13	9	Gasket for oil pump	2	
			14	11	Cheese head screw M6x25	6	
			15	51	Flat Gasket for oil sump	2	
			16	42	Cheese head screw DIN6912-M6x20	24	
			17	45	Washer	8	
			18	37	Washer	16	
			19	12	Connecting Rod	4	
			20	38	Screw M4x0.7x55	16	
			21	22	O-Ring ISO3601-18x1.5	2	
			22	61	Rivet	2	
2	913 100 920 2	Hose & clamp Kit	1	26	Intake hose	4	517143990102
			2	25	Hose clamps	8	
3	913 100 940 2	Oil Sump Kit	1	53	Oil Sump	1	517143990103
			2	52	Separator	1	
			3	51	Gasket	1	
			4	32	Screw plug	1	
			5	31	Sealing ring	1	
			6	42	Cheese head screw DIN6912-M6x20	10	
4	913 100 941 2	Oil filter & Pressure relief valve	1	50	Oil filter	1	517143990104
			2	42	Cheese head screw DIN6912-M6x20	10	
			3	51	Gasket	1	
			4	49	Pressure Relief valve	1	
5	913 100 924 2	Damper Kit	1	54	Damper	1	517143990105
			2	56	Nut M10X1.25	2	
6	913 100 942 2	Bracket kit	1	55	Bracket	1	517143990106
			2	56	Nut M10X1.25	1	
7	913 100 932 2	Liner Kit	1	15	Crankcase liner	1	517143990107
			2	14	Liner O-Ring	1	
			3	16	Inlet Valve	1	
			4	18	Gasket for Head	1	
			5	21	Hexagon Flange Bolt M8X100	4	
			6	20	Cheese head screw ISO14580-M6x35	1	
			7	45	Washer	2	
			8	37	Washer	4	
			9	38	Screw M4x0.7x55	4	
8	913 100 925 2	Left Oil Pump Kit	1	40	Oil Pump	1	517143990108
			2	9	Gasket for oil pump	1	
			3	11	Cheese head screw M6x25	3	
9	913 100 926 2	Right Oil Pump Kit	1	10	Oil Pump	1	517143990109
			2	9	Gasket for oil pump	1	
			3	11	Cheese head screw M6x25	3	
10	913 100 933 2	Head Assy Kit	1	16	Inlet Valve	1	517143990110
			2	18	Gasket for Head	1	
			3	17	Valve Plate Assy	1	
			4	19	Cylinder Cover Machining	1	
			5	21	Hexagon Flange Bolt M8X100	4	
			6	20	Cheese head screw ISO14580-M6x35	1	
			7	38	Screw M4x0.7x55	4	
			8	37	Washer	4	
			9	45	Washer	2	
ⓓ 11	913 100 947 2	Oil Pressure Switch Kit	1	33	Oil Pressure Switch	1	517143990141
			2	27	Cable tie	1	
ⓓ 12	913 100 943 2	Resonator Kit	1	23	Resonator	1	517143990112
			2	24	Senter Filter sub assembly	2	
			3	22	O ring	2	
			4	26	Hose_cover with pin tank	4	
			5	25	Clamp	8	
			6	27	Cable tie	3	
13	913 100 928 2	Discharge Pipe Kit	1	47	Discharge pipe assy	2	517143990113
			2	48	Discharge pipe assy	2	
			3	46	Fitting	8	
			4	41	Discharge tank	1	
			5	42	Cheese head screw DIN6912-M6x20	4	
			6	44	Plug	1	
			7	43	Washer	1	
14	913 100 935 2	Intake Filter Kit	1	24	Senter Filter sub assembly	2	517143990114
			2	22	O ring	2	
			3	27	Cable Tie	3	
15	913 100 929 2	Fan Kit	1	36	Fan Assy	1	517143990115
			2	37	Washer	16	
			3	38	Screw M4x0.7x55	16	
			4	27	Cable tie	4	
16	913 100 936 2	Piston kit	1	13	Piston	1	517143990116
			2	16	Inlet Valve	1	
			3	18	Gasket for Head	1	
			4	21	Hexagon Flange Bolt M8X100	4	
			5	20	Cheese head screw ISO14580-M6x35	1	
			6	45	Washer	2	
			7	37	Washer	4	
			8	38	Screw M4x0.7x55	4	
			9	12	Connecting Rod	1	
			10	51	Gasket for Oil sump	1	
			11	42	Cheese head screw DIN6912-M6x20	10	
			12	14	Liner O ring	1	
17	9131009392	Drain Plug-Fitting	1	31	Sealing Ring DIN7603-A18x22	1	517143990117
			2	32	Hexagon Head Screw Plug M18x1.5	1	
18	9131009002	Crankcase	1	2	O-Ring ISO3601-94x2.5	1	517143990118
			2	42	Cheese head screw ISO4762-M6x20	10	
			3	11	Cheese head screw ISO4762-M6x25	3	
			4	34	Washer	1	
			5	35	Screw Plug	1	
			6	31	Sealing Ring DIN7603-A18x22	2	
			7	8	Cheese Head Screw	3	
			8	56	Locking nut	2	
			9	32	Hexagon Head Screw Plug M18x1.5	1	
			10	14	O-Ring	2	
			11	51	Gasket	1	
			12	7	Crankcase	1	
			13	9	Flat Gasket	1	
19	Buy Out Part	Primary Filter Kit	1	57	Primary Filter	1	517143990119
20	Buy out Part	Secondary Filter Kit	2	58	Secondary Filter	1	517143990120

Do not scale the drawing.	Unless otherwise stated. 1. All dimns in mm. 2. Drg drawn to first angle projection. 3. For supplier drg release, refer angle of projection on supplier drg.	Nominal Dimension	Tolerance in µm(microns) Refer IS919-2
If in doubt, ask.		-	-

Date	Mod. No.	Sl No.	Zone	Modification	Chkd	Appd
12/12 2024	-	g	-	4Cy eCompressor changes -eComp without elbow.	APA	PRA
18/11 2024	22CU V40 014	f	-	E COMPRESSOR ASSY. UPDATED WITH CHILD PART LOCALIZATION: 1. RESONATOR 2. OIL SUMP 3. DISCHARGE PIPE 4.CY. COVER 5.DISCHARGE TANK 6.C CLASS PARTS 7.CYLINDER LINER 8.CRANK SHAFT 9.HOLLOW SCREW 10.HOSE BELLOW 11.WASHER & 12.CASTINGS 13.OIL PUMP	APA	PRA
04/12 2023	15PM 440 109	e	L8	ELECTRICAL CONNECTOR DETAILS ADDED	APA	MVS
29/09 2023	15PM 440 104	d	G7 - -	OIL PRESSURE SWITCH KIT 517143990141 WAS 517143990111. REPEAT ITEM 'INTAKE HOSE' QTY. 4 REMOVED FROM RESONATOR KIT EXPLODED VIEW UPDATED PART NUMBER COLUMN REMOVED	APA	MVS
10/09 2022	12PP 287 054	c	-	CAD UPDATED	APA	MVS
27/08 2022	15PM 440 069	b	-	SUPPLIER KIT DETAILS ADDED	APA	MVS
17/06 2022	14PM 222 068	a	-	CAD UPDATED	APA	MVS
19/12 2020	20PM 312 000	NR	--	NEW RELEASE	--	--

For Bar Codes System of Supplier Parts, refer TS 11879
For Numbering of Aggregates Manufactured in-house, refer TS 11842
For List of Parts having Recall & Traceability Requirements, refer TS 10814

For all Vehicle Components & Assemblies.
Tata Group Mark & Part No. to be marked as per TS 10806

For M1 & N1 class of vehicles, applicable standards are:
TS11414,TS11418,TS11419 & TS11420

Material description,size,spec,std No.

Proprietary Part: Refer Supplier Drawing

Classification of characteristics:
- ● *Critical,* ▲ Key Instruction
- ◐ *Major,* *(None) Minor*
- ● *Applies to upper limit*
- ● *Applies to lower limit*

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2024	SIGN	DATE
Drawn	RKG	13/12
Chkd	APA	13/12
Appd	PRA	13/12

Scale: 1:1

Envelope Dimensions 356.07 x 389.82 x 622.00

Final mass in kg. 51.460

Surface protection as per standard -

Tolerance as per standard -

Product/Group No. 9/12 EV PMeSEVA BUS.

Replaces Drg -

Reference Drg No. -

Opposite Hand Drg/Part No. -

TATA MOTORS LIMITED
ERC - PUNE

Drg/Part Designation

AIR COMPRESSOR
-

Drg/Part No.

5171 4370 01 01

Sheet No....2.... of ....2.... Sheets

For explanation of drawing format contents, refer to TS11103
8845 4130 70 66 PCF- 071(R4) 6/2011

Original drawing size A0